Writing Inequalities from Verbal Constraints
Answer Key

Grade 6–7 Expressions and Equations

Quick Answers

1. 48, ∞, *hiopen*
2. $-\infty$, 14, *loopen*
3. $-\infty$, 11, *loopen*, 11
4. $-\infty$, 10, *loopen*
5. 20, ∞, *hiopen*, 20
6. $-\infty$, 43, *loopen*
7. $-\infty$, 10, ¡built-in function open¿
8. $-\infty$, 12, *loopen*
9. $-\infty$, 15, *loopen*
10. —

Curriculum

Level: Grade 6–7 Expressions and Equations
7.EE.B.4 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–4 of 5 across 12 tagged checks.

1. Ride height: at least 48 inches.

Step 1. Name the variable in words: h is a rider's height in inches.
Step 2. "At least" means the boundary counts — a rider who is exactly 48 inches tall may ride — so the symbol is $\geq$, not $>$.

$$\boxed{h \geq 48}$$

Step 3. On the number line: a *closed* circle at 48 with the shading running to the right.
Common wrong answer: $h > 48$. That would turn away a rider who is exactly at the line, which is not what the sign says.

2. Van capacity: no more than 14 passengers.

Step 1. p is the number of passengers on a trip.
Step 2. "No more than" is the same as "at most": 14 passengers is allowed, 15 is not, so the symbol is $\leq$.

$$\boxed{p \leq 14}$$

Step 3. Closed circle at 14, shading to the left.
Teaching note: in context p also cannot be negative, so a fussy answer is $0 \leq p \leq 14$. Accept $p \leq 14$ — the sheet asks for the constraint the sign states.

3. Maya's fair budget.

Step 1. s is the number of snacks. Entry is a one-time \$16; each snack adds \$4, so the total spent is $16 + 4s$ dollars.
Step 2. "Does not overspend" means the total is at most her \$60:
$16 + 4s \leq 60$
Step 3. Subtract 16 from both sides: $4s \leq 44$. Divide by 4 (a positive number, so the symbol does not flip).

$$\boxed{s \leq 11}$$

Step 4. Check the boundary: $16 + 4(11) = 60$ exactly ✓ Closed circle at 11, shading left.
Common wrong answer: $4s + 16 \leq 60$ solved as $s \leq 15$ — the \$16 was subtracted from $4s$ instead of from 60.

4. Gym membership within \$150.

Step 1. m is the number of months. The joining fee is paid once, so the total cost is $30 + 12m$ dollars — the 30 is not multiplied by m.
Step 2. "At most \$150" gives $30 + 12m \leq 150$.
Step 3. Subtract 30: $12m \leq 120$; divide by 12.

$$\boxed{m \leq 10}$$

Step 4. Check: $30 + 12(10) = 150$ ✓ Closed circle at 10, shading left.
Common wrong answer: writing $30m + 12 \leq 150$ — the one-time fee and the repeating charge were swapped. Ask which number happens again every month.

5. Fundraiser: at least \$500.

Step 1. w is the number of wreaths sold. Money raised is $180 + 16w$ dollars.
Step 2. "At least \$500" points the other way from the last two problems — they need to reach or pass the goal:
$180 + 16w \geq 500$
Step 3. Subtract 180: $16w \geq 320$; divide by 16.

$$\boxed{w \geq 20}$$

Step 4. Check the boundary: $180 + 16(20) = 500$ ✓ Closed circle at 20, shading to the *right*.
Listen for: shading the wrong way. Test one value: 25 wreaths gives \$580, which does reach the goal, so the shading covers 25.

6. Elevator load limit.

Step 1. b is the number of boxes. Total weight is $165 + 45b$ pounds.
Step 2. "No more than 2100 pounds" gives $165 + 45b \leq 2100$.
Step 3. Subtract 165: $45b \leq 1935$; divide by 45.

$$\boxed{b \leq 43}$$

Step 4. Read it back. Boxes come in whole numbers, so the operator may load any whole number of boxes up to and including 43. Check: $165 + 45(43) = 2100$ exactly, so 43 boxes is right at the limit and is allowed.
Teaching note: this is the step students skip — the algebra ends at $b \leq 43$, but the *answer to the question asked* is "at most 43 boxes".

7. Drone above 90 metres.

Step 1. t is seconds since the descent began. Height after t seconds is $240 - 15t$ metres — descending, so the term is subtracted.
Step 2. "Above 90 metres" is strict, so $240 - 15t > 90$.
Step 3. Subtract 240 from both sides: $-15t > -150$.
Step 4. Divide both sides by -15. Dividing by a negative number *reverses* the symbol:

$$\boxed{t < 10}$$

Step 5. Check with a value: at $t = 4$ the height is $240 - 60 = 180$ metres, which is above 90 ✓ At $t = 12$ it is 60 metres, which is not. Open circle at 10 (at exactly 10 seconds the drone is *at* 90 metres, not above it), shading to the left.
Common wrong answer: $t > 10$ — the flip was forgotten. The check value catches it instantly, which is why it is worth doing every time.

8. What Jonah got wrong.

Step 1. "No more than 12" means 12 is the largest value still allowed. Jonah's symbol, $<$, excludes 12, so his inequality throws away a value the sentence permits.
Step 2. The phrase calls for the symbol that includes the boundary, $\leq$:

$$\boxed{n \leq 12}$$

Step 3. On the number line the difference is visible: Jonah's version has an *open* circle at 12; the correct version has a *closed* one, with the shading running left in both.
Full credit on the sentence: it has to say that 12 itself is allowed (or that "no more than" includes the boundary). "He used the wrong sign" has spotted the error without explaining it.

9. Rectangle with perimeter at most 46 cm.

Step 1. l is the length in centimetres. Perimeter is two widths plus two lengths: $2(8) + 2l$, which is $16 + 2l$ centimetres.
Step 2. "At most 46" gives $2(8) + 2l \leq 46$.
Step 3. Simplify and subtract 16: $2l \leq 30$; divide by 2.

$$\boxed{l \leq 15}$$

Step 4. Check the boundary: a 8 by 15 rectangle has perimeter $16 + 30 = 46$ ✓ Closed circle at 15, shading left.
Teaching note: a length also has to be positive, so the full set of usable lengths is $0 < l \leq 15$. Worth raising once the inequality is written — the context, not the algebra, supplies that second constraint.

10. Writing a story for $25 + 8n \geq 65$. *Open response — many correct answers, flagged for manual review. A model answer follows.*

Step 1. Read the structure first. There is a one-time 25, a repeated 8 attached to n, a target of 65, and "at least" pointing at $\geq$.
Step 2. A story that fits: Priya already has \$25 saved and adds \$8 to her jar each week. She needs at least \$65 for a bike helmet. Here n counts the weeks she keeps saving, 25 is what she starts with, 8 is what she adds per week, and 65 is the goal.
Step 3. Solve it: $8n \geq 40$, so $n \geq 5$.
Step 4. Read it back into the story: after 5 weeks she has exactly $25 + 40 = \$65$, so 5 weeks is enough, and every week after that is too. The graph is a closed circle at 5 shading right.
Full credit: the one-time amount and the repeating amount are matched to the right places, n is named as a count, and the interpretation says "5 or more" rather than just restating the symbols. Any story with that shape earns the marks.